

Generating Sets and Cayley graphs

Section 7

Want to understand non-cyclic groups, requiring multiple **generators**.

Recall: For $a \in G$, $\langle a \rangle$ is the smallest subgroup of G containing a .

$$\{ \dots, a^{-2}, a^{-1}, e = a^0, a, a^2, a^3, \dots \}$$

What should $\langle a, b \rangle$ be?

$$\{ a^{i_1} b^{i_2} a^{i_3} b^{i_4} \dots a^{i_k}, e, a, b, a^n, b^m, ab, ba, a^n b^m, b^m a^n \mid i_1, \dots, i_k \in \mathbb{Z}, n, m \in \mathbb{Z} \}$$

Prop: For $a, b \in G$, $\langle a, b \rangle \leq G$.

Pf:

① Closed:
$$\left(a^{i_1} b^{i_2} \dots a^{i_k} \right) \left(a^{j_1} b^{j_2} \dots a^{j_l} \right)$$
$$= a^{i_1} b^{i_2} a^{i_k + j_1} b^{j_2} \dots a^{j_l} \in \langle a, b \rangle.$$

② $e \in \langle a, b \rangle$: $e = a^0 = b^0 \in \langle a, b \rangle$

③ Inverses: let $g = a^{i_1} b^{i_2} \dots a^{i_k} \in \langle a, b \rangle$

$$g^{-1} = a^{-i_k} \dots b^{-i_2} a^{-i_1} \in \langle a, b \rangle$$

$$\left(a^{i_1} b^{i_2} \dots a^{i_k} \right) \left(a^{-i_k} \dots b^{-i_2} a^{-i_1} \right) = e.$$

$$(xy)^{-1} = y^{-1}x^{-1}$$

$$(xyz)^{-1} = z^{-1}y^{-1}x^{-1}$$

□

For arbitrary sets $S \subseteq G$, let's be a bit more formal.

Def: Let $\{S_i \mid i \in I\}$ be an indexed collection of sets.

$\bigcap_{i \in I} S_i = \{x \mid x \in S_i, \forall i \in I\}$.

each S_i is a set

$$\begin{array}{ll} S_a, S_b, S_c & I = \{a, b, c\} \\ \bigcap_{i \in I} S_i = S_a \cap S_b \cap S_c. \end{array}$$

Thm: Suppose $H_i \leq G$ for $i \in I$. Then $K = \bigcap_{i \in I} H_i \leq G$.

Pf:

① Closed: let $x \in K, y \in K$. Then $x, y \in H_i$ for all $i \in I$, so since $H_i \leq G$, we have $x * y \in H_i, \forall i \in I$. Thus $x * y \in K$.

② Identity: $e \in H_i, \forall i \in I$ since $H_i \leq G$. Thus $e \in K$.

③ Inverses: if $x \in K$ then $x \in H_i, \forall i$, so $x^{-1} \in H_i, \forall i$
so $x^{-1} \in K. \square$

Special case:
if $H_1, H_2 \leq G$
then
 $H_1 \cap H_2 \leq G$.
 $I = \{1, 2\}$.

Example: $G = \mathbb{Z}$

$$2\mathbb{Z} \cap 3\mathbb{Z} = \{ -6, 0, 6, 12, \dots \} = 6\mathbb{Z}$$

$$\begin{array}{ccc} \text{"} & & \text{"} \\ \{ \dots, -4, -2, 0, 2, 4, 6, \dots \} & \cap & \{ \dots, 0, 3, 6, \dots \} \end{array}$$

$$\overset{2 \cdot 2}{4}\mathbb{Z} \cap \overset{2 \cdot 3}{6}\mathbb{Z} = \overset{2 \cdot 2 \cdot 3}{12}\mathbb{Z}$$

$$a\mathbb{Z} \cap b\mathbb{Z} = \text{lcm}(a, b) \cdot \mathbb{Z}$$

↑ least common multiple

Def: Let G be a group and $a_i \in G$ for all $i \in I$. Then the subgroup generated by $\{a_i \mid i \in I\}$ is:

$$\langle a_i \mid i \in I \rangle = \bigcap_{\substack{H \leq G \\ a_i \in H, \forall i \in I}} H$$

This is the smallest subgroup of G containing all $a_i, i \in I$.

Def: $\{a_i \mid i \in I\}$ generates G if

$$\langle a_i \mid i \in I \rangle = G.$$

Def: G is finitely generated if there exists a finite set $a_1, \dots, a_n$ such that $\langle a_1, \dots, a_n \rangle = G$.

$$\langle a_i \mid i \in \{1, \dots, n\} \rangle$$

Example: Any cyclic group is finitely generated.

$G = \langle a \rangle$ can be generated by one element.

Example: $\{2, 3\}$ in $\mathbb{Z}_6$.

$$\langle 2 \rangle = \{0, 2, 4\}$$

$$\langle 3 \rangle = \{0, 3\}$$

$$\langle 2, 3 \rangle = \{0, 2, 4, 3, 5, 1\}$$

Example: $G = V_4$.

$$G = \langle a, b \rangle = \{e, a, b, c\}$$

$$\langle a, b, c \rangle$$

V_4	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Example: $G = \langle \mathbb{R}, + \rangle$.

Not finitely generated, since any finitely generated group must be countable.

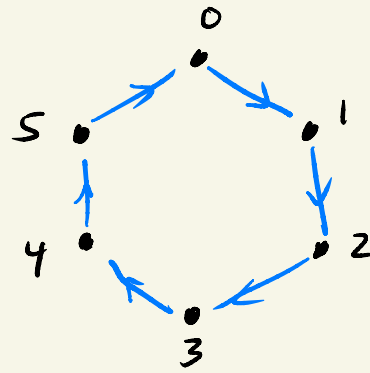
Cayley (di)graphs

We can depict a finite group G with generating set S via a Cayley graph.

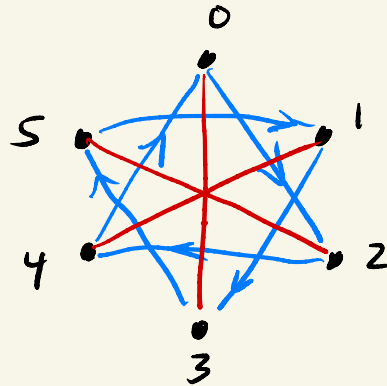
- vertices correspond to elements of G
- edges $x \xrightarrow{s} y$ for $s \in S$ mean $y = xs$.

The resulting graph is denoted $\text{Cay}_R(G, S)$.

Example: $G = \mathbb{Z}_6$, $S = \{1\}$



Example: $G = \mathbb{Z}_6$, $S = \{2, 3\}$



Properties of Cayley graphs

1. The digraph is connected:
you can go from any g to
any h by following edges

• a path from e to g
gives an expression
 $g = s_1 \cdots s_k$, $s_i \in S, \forall i$.

2. At most one edge goes
from g to h .

• There is only one element x
with $gx = h$.

3. Each vertex has one incoming
edge and one outgoing edge
for each $s \in S$

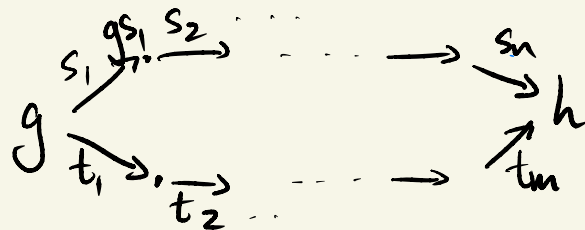
• For each $s \in S$ and $g \in G$
we have

$$gs^{-1} \xrightarrow{s} g \xrightarrow{s} gs$$

$$gs^{-1} \cdot s = g$$

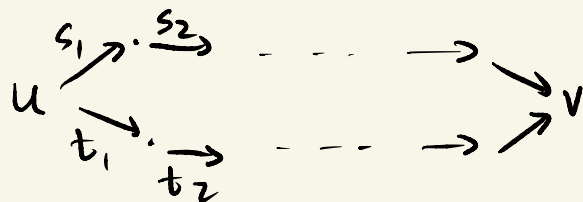
4.

If



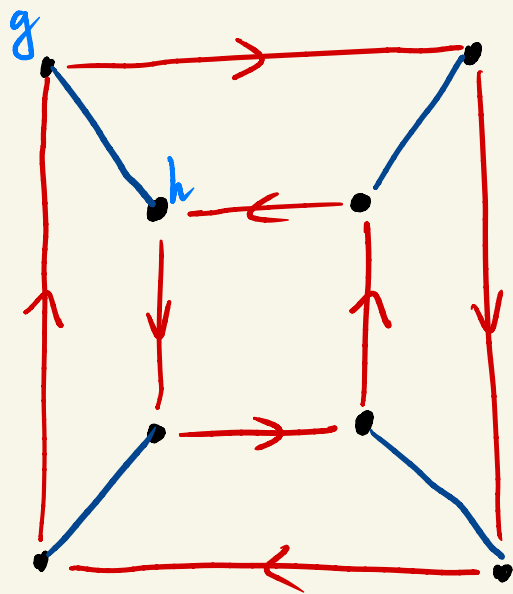
means $gs_1s_2\cdots s_n = h$
 $gt_1t_2\cdots t_m = h$ so
 $s_1\cdots s_n = t_1\cdots t_m$.

then



Prop: Any digraph satisfying Properties (1)–(4)
 is the Cayley graph for some generating set of
 some group.

Pf: Book.



1. Connected

2. Only one edge $g \rightarrow h$

3. Each vertex has 1 edge of each kind

4. Path agreements

$$b = r b r$$

$$r^4 = e$$

$$r b \neq b r$$

$$b^2 = e$$

